



FASCIA SCIENCE AND CLINICAL APPLICATIONS – CLINICAL APPROACHES: REVIEW

Selected fascial aspects of osteopathic practice

Paolo Tozzi, Bsc (Hons) Ost, DO, PT ^{a,b,c,*}

^a *Centro di Ricerche Olistiche per la Medicina Osteopatica e Naturale, C.R.O.M.O.N., Via Pasquale Fiore 18, Rome, Italy¹*

^b *Osteopathie Schule Deutschland, Mexikoring 19, D-22297 Hamburg, Germany*

^c *Dresden International University, Freiburger Str. 37, D-01067 Dresden, Germany*

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Summary Fascia is a connective tissue organised as a three-dimensional network that surrounds, supports, suspends, protects, connects and divides muscular, skeletal and visceral components of the body. Studies suggest that fascia reorganises itself along the lines of tension imposed or expressed in the body, and in ways that may cause repercussions to fascial restriction that are body-wide. This may potentially create stress on any structures enveloped by fascia itself, with consequent mechanical and physiological effects. From an osteopathic perspective, fascial techniques aim to release such tensions, decrease pain and restore function. The proposed mechanism for fascial techniques is based on various studies that have looked at the plastic, viscoelastic and piezoelectric properties of connective tissue. This review explores some of the features described above, together with evidence supporting the therapeutic efficacy of fascial manipulation, offering a selected overview of the fascial component in osteopathic assessment and treatment.

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Recent fascial insights

Fascial contributions to biomechanics

The ubiquitous distribution of fascia permeates the human body, forming a continuous matrix of structural support,

serving different functions. It has been traditionally considered as an inert structure, with passive roles such as being a cushioning system, providing muscular attachments and investing different body structures at various depths (Standring, 2004; Williams, 1995), being generally considered less important than the tissues with which it is

* Via Feste Avieno 150, 00136 Rome, Italy. Tel.: +39 3486981064 (Cell); fax: +39 06 97749900.

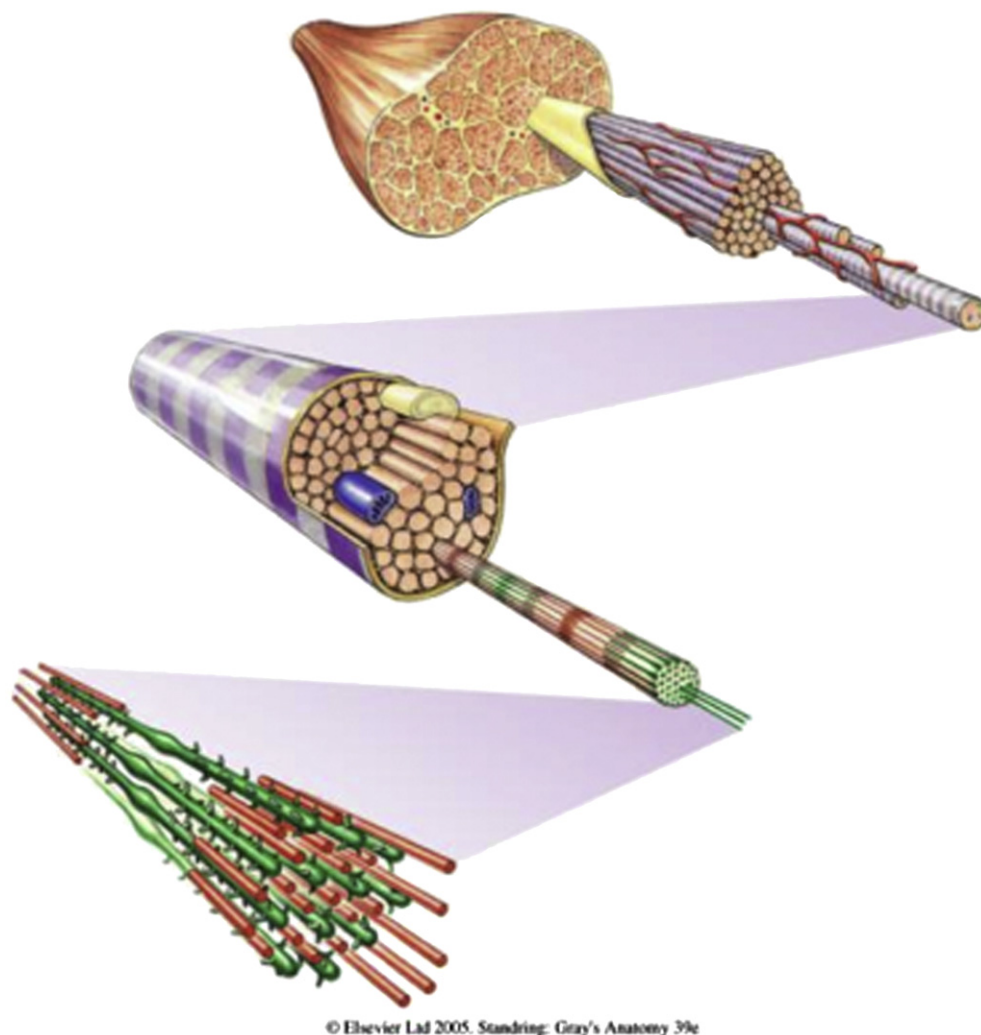
E-mail address: pt_osteopathy@yahoo.it.

¹ www.scuoladiosteopatia.it

associated (Benjamin, 2009). However, it appears to be a very active tissue, fundamental to the economy of the body and its health (Snyder, 1956), playing different physiological and functional roles that are still poorly understood in their complexity, such as those related to joint stability, general movement coordination, proprioception and nociception. It's potential role in many pathologies such as back pain, as well as in wound healing and tissue repair also requires further investigation.

A primary biomechanical role for the Thoraco–Lumbar Fascia (TLF) has been demonstrated for the stability and mobility of the lumbar spine (Bogduk and Macintosh, 1984; Gracovetsky et al., 1981) together with its role in determining a 'force closure' system that transfers force and stabilises the pelvic girdle, while distributing movement to the hips and lower limbs (Vleeming et al., 1995). This supports the idea that fascial structures do not work in an isolated manner, instead seeming to be integrated into a more complex 'connective tissue system'. Following the intra-abdominal pressure model previously proposed by Bartelink (1957), Gracovetsky et al. (1985) referred to a 'posterior ligamentous system' as the main structure responsible for lumbar spine extension during lifting. This

system has been found in anatomical and functional inter-relations (Willard, 1997), to be composed of the capsules of the zygapophyseal joints, the interspinous and supraspinous ligaments, together with the posterior layer of the TLF. In addition a 'hydraulic amplifier' effect (Gracovetsky et al., 1977) of the TLF itself may contribute to spinal extension during lifting: in fact, by surrounding the back muscles as a retinaculum, the TLF could serve to brace these muscles and enhance their power by up to 30% (Hukins et al., 1990). To support this hypothesis, EMG based measurements have shown a strong tensional load-bearing function of dorsal fascial tissues during healthy forward bending of the trunk (Shirado et al., 1995). Conversely, this load shifting has been found to be absent in LBP patients. Furthermore, this function of force transmission seems to occur at both inter and intra muscular levels: Huijing (2009) showed epimysial connections between both synergistic and antagonist muscles, via myofascial force transmission between and within muscles (Fig. 1). Fascia dissections have also revealed a fibre distribution relating to precise motor vectors (Stecco et al., 2009). This arrangement of connective tissue around the muscular tissue has been hypothesised to have a specific role on force generation and



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Figure 1 Basic fascial structures surrounding a skeletal muscle are shown together with its different levels of organisation, from whole muscle to fasciculi, single fibres, myofibrils and myofilaments. Reproduced from Standing (2004) with permission.

transfer: as muscles actively contract to generate force, they pass this force outwardly by deforming the surrounding connective tissues (Brown et al., 2011), based on the magnitude and direction of force application. Activation of such a muscular complex through a connective tissue complex, creates a transfer of energy and a coupling motion of the limbs, **for instance, from the upper extremity through the spine and into the lower limbs during gait, as an integrated system** (Vleeming et al., 1995) (Fig. 2). It also

helps to tighten the connective tissue support structures, resulting in a stabilising action to the spine (Snijders et al., 1993), as well as working functionally as a transmission belt, an active storage and release of elastic anti-gravitational energy (Dorman et al., 1993). In fact, recent ultrasound (US) based measurements have indicated that fascial tissues are commonly used for dynamic energy storage during oscillatory movements such as walking (Fukunaga et al., 2002). During movements, the supporting

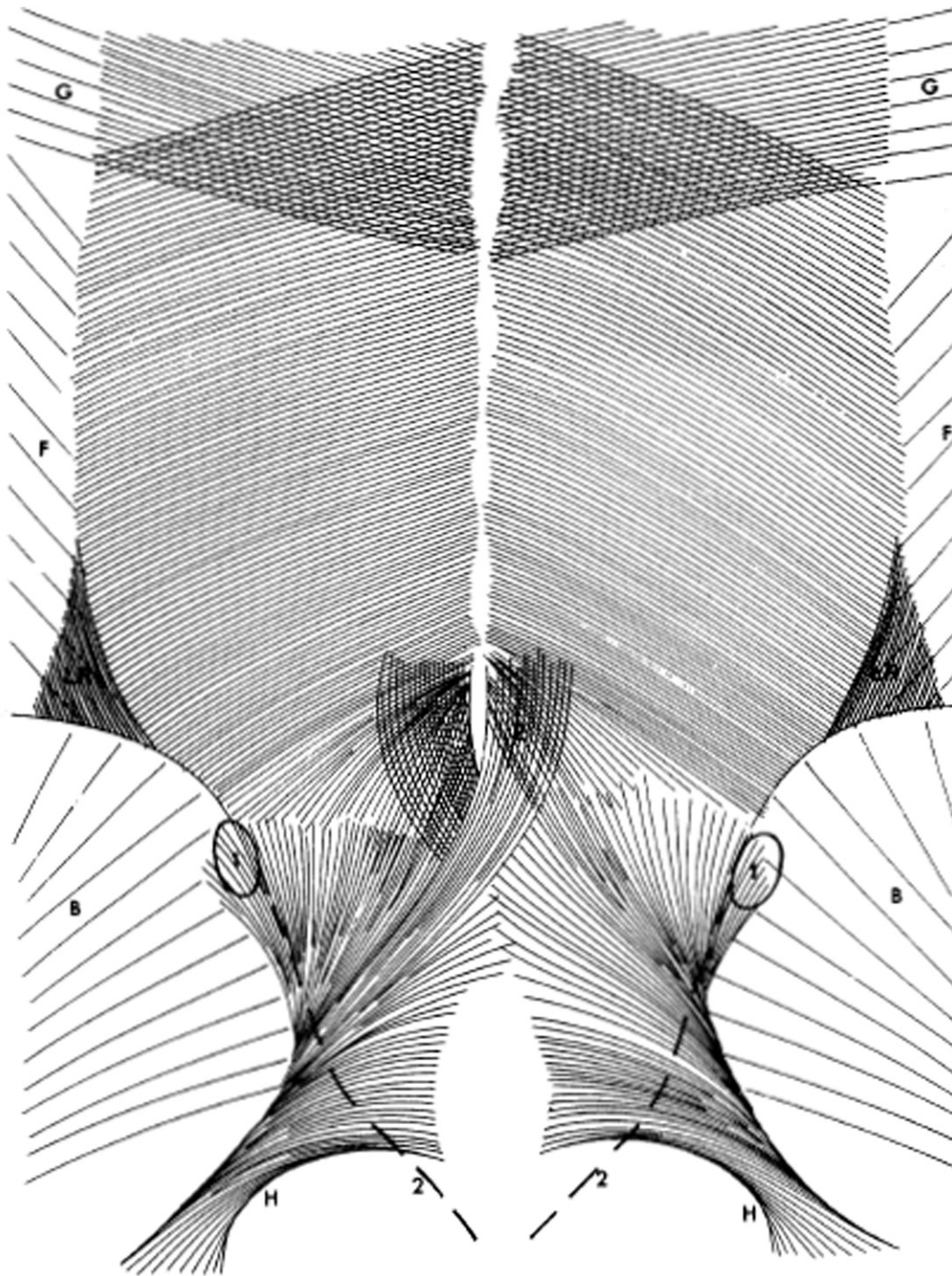


Figure 2 The deep layer of the thoracolumbar fascia and its attachments to: (B) gluteus medius (E) attachments between the deep layer and the erector spinae muscle (F) the internal oblique (G) serratus posterior inferior (H) the sacrotuberous ligament (1) PSIS (2) sacrum. Reproduced from Vleeming et al. (1995) with permission.

skeletal muscles contract more isometrically, while the loaded fascial elements lengthen and shorten like elastic springs (Fig. 3). By applying this concept at a functional level, the concerted action by the abdominals, gluteals, erector spinae and latissimus dorsi to increase or maintain tension in the TLF during lifting (Adams and Dolan, 1997) suggests that the TLF may not only play a mechanical function to transfer forces along its muscular attachments,

but also have a proprioceptive function to coordinate muscle group action.

Fascia as sensory organ

The hypothesis that fascia may play an important role in proprioception, especially dynamic proprioception, and that an impairment of such function can be related to back pain, has been supported by the findings of both free nerve endings and encapsulated receptors in the fascia (Fig. 4), Ruffini and Pacini corpuscles in particular (Schleip, 2003a; Stecco et al., 2010). Numerous studies have shown that individuals with LBP have altered, usually decreased, lumbosacral proprioception in different postures compared to healthy control subjects (Brumagne et al., 2000; O'Sullivan et al., 2003). Panjabi's explanatory model of LBP injuries suggests that a single trauma or cumulative microtrauma cause sub failure injuries of paraspinal connective tissues and their embedded mechano receptors (Panjabi, 2006). Following a mechanical stimulation beyond their physiological limit, these receptors have the ability to become nociceptors (Stecco et al., 2007). This series of events may lead to altered mechanoreceptor feedback, consequent connective tissue alterations and neural adaptations (Panjabi, 2006). As shown in surgical examinations of the posterior layer of the TLF, people with LBP revealed frequent signs of injury and of inflammation (Bednar et al., 1995; Dittrich, 1964), suggesting that the same fascia could be prone to sub failure injuries and to a pain generative role (Schleip et al., 2007).

Initial investigations on TLF innervation show that it may indeed be a source of pain as it is appropriately innervated (Stilwell, 1956, 1957), especially at its attachment to the supraspinous ligaments (Yahia and Newman, 1993). In rats (Hoheisel et al., 2011) as well as in humans (Yahia et al., 1992) the dense innervation of the TLF with afferent free nerve endings, including nociceptive ones, may suggest its potential role as a source of nociceptive input. Irritation of these primary afferent nociceptive fibres is capable of initiating the release of neuropeptides, which in turn may change the normal tissue texture of the surrounding

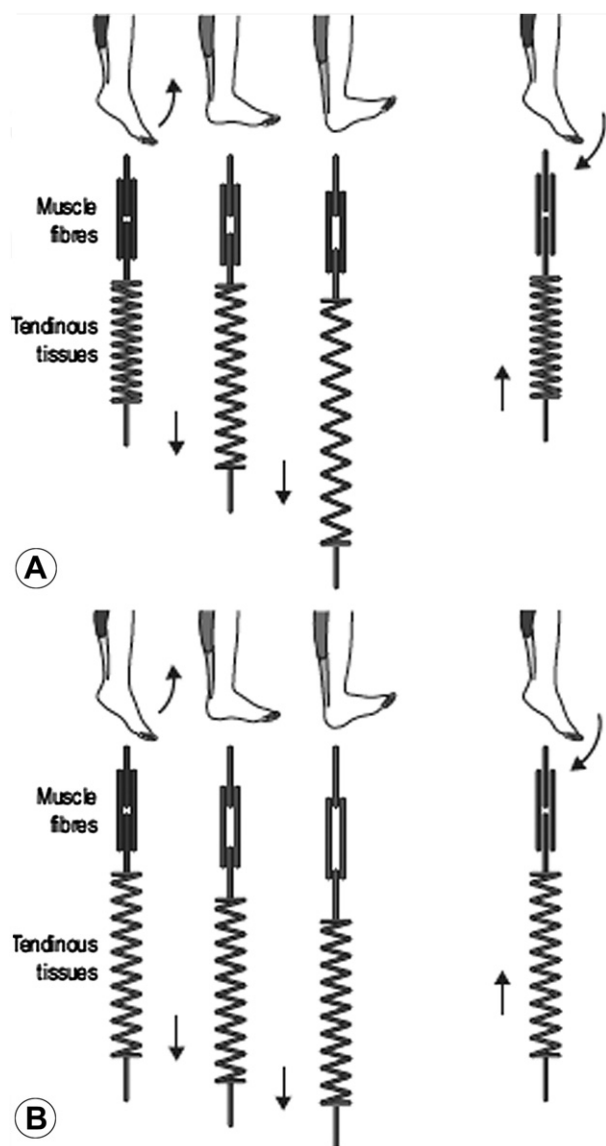


Figure 3 Length changes of fascial elements and muscle fibres in an oscillatory movement with elastic recoil properties (A) and in conventional muscle training (B). The elastic tendinous (or fascial) elements are shown as springs, the myo-fibres as straight lines above. Note that during a conventional movement (B) the fascial elements do not change their length significantly while the muscle fibres clearly change their length. During movements like hopping or jumping however the muscle fibres contract almost isometrically while the fascial elements lengthen and shorten like an elastic yoyo spring. Illustration adapted from Kawakami et al. 2002. Reproduced from Mueller and Schleip (2012) with permission.

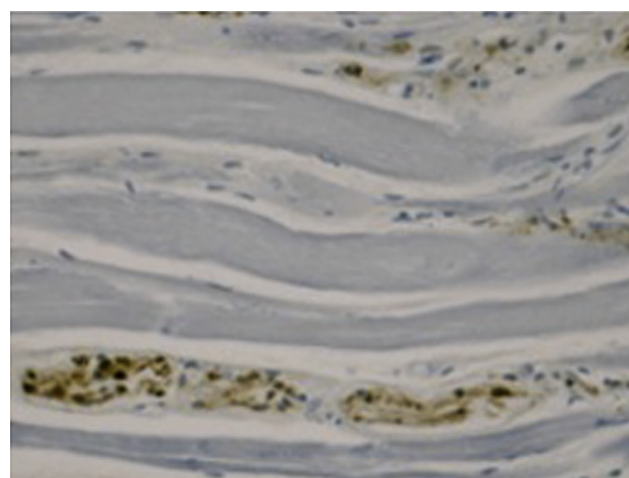


Figure 4 Nerve fibres among the collagen fibres of the antibrachial fascia (anti S100 immuno histochemical stain, 200×) Reproduced from Stecco et al. (2008) with permission.

connective tissue, through their interaction with fibroblasts, mast cells, and immune cells (Levine et al., 1993). The nerve fascia, such as the epineurium and perineurium, are innervated by 'nervi nervorum', which can generate neurogenic inflammation, evoke nociception, and possibly nerve trunk pain (Bove, 2008). A vicious cycle may take place at this stage: pain-related fear leads to decreased movement, resulting in connective tissue remodelling, followed by inflammation, nervous system sensitisation and further decreased mobility (Langevin and Sherman, 2007). The resultant cascade of events is thought to prime the development of pain (Weidenbaum and Farcy, 1990) and to cause the evolution of persistent pain (Melzack et al., 2001), to eventually establish an adaptive response of the whole organism (Willard, 1995).

Contractile properties of fascia

The roles of the fascia in force transmission, movement coordination, proprioception, and nociception may not be fully understood without taking into account the recently discovered property of fascial contractility. The presence of myofibroblasts in the connective tissue has been demonstrated to relate to wound healing as well as in both normal and pathological contractile tissue processes (Gabbiani, 1998). Furthermore, investigation of myofibroblasts' structure has demonstrated the presence of alpha smooth muscle actin as well as of specific adherent junctions (Hinz et al., 2004), supporting a plausible capacity of fascial contractility. An increased density of myofibroblast found in human lumbar fascia not only suggested its ability to actively contract in a smooth muscle-like manner, but also its potential influence on musculoskeletal dynamics (Schleip et al., 2005) as well as on resting muscle tone (Klingler et al., 2007). In addition, electron photo microscopic studies of the human fascia have shown smooth muscle cells embedded in the collagen fibres, including intrafascial capillaries and autonomic nerves (Staubesand and Li, 1997). Similar evidence is available with regards to the human lumbo dorsal fascia (Yahia et al., 1993), supporting the hypothesis that autonomic nerve fibres may regulate a fascial pre-tension via the smooth muscle cells contractility, independently from muscular tonus (Staubesand and Li, 1996). Autonomic nerve fibres may also exert an indirect effect on proprioception by decreasing the blood flow to skeletal muscles (Thomas and Segal, 2004) as well as impairing proprioception by the action exerted on muscle spindle receptors: either by decreasing their sensitivity to muscle length changes (Roatta et al., 2002), or by affecting their basal discharge rate (Hellström et al., 2005). Note that these effects have been demonstrated mostly in animal studies.

Lastly, sympathetic fibres may play a role in pain modulation, by activating sensitised primary afferent fibres either directly or indirectly (Roberts and Kramis, 1990), suggesting that they may be involved in the genesis or maintenance of pain states in the connective tissue (Basbaum and Levine, 1991). However, autonomic supply may not be the only influence on smooth-muscle cells contractility in fascia. An elevated pH may, in fact, alter the general fascial tone, producing smooth muscle contraction and even spasm (Schleip, 2003b). Such

a condition may occur in relation to respiratory alkalosis resulting from a breathing pattern disorders (Lum, 1987) leading to profound vascular implications (Nakao et al., 1997). Conversely, a more acidic environment seems to exert a modulating action on connective tissue cells' metabolism (Ohshima and Urban, 1992) as well as on the balance between synthesis and degradation of the matrix (Melrose and Ghosh, 1988).

Osteopathic fascial manipulation (OFM)

Somatic dysfunction of the myofascial system

Since the origins of the osteopathic profession, the role of fascia has been considered as an important focus in achievement of optimal therapeutic outcomes (Lee, 2006): *"Fascia is the place to look for the cause of disease and the place to consult and begin the action of remedies in all diseases...By its action we live and by its failure we die...The fascia is the ground in which all causes of death do the destruction of life"* (Still, 1902). Osteopathic medicine uses manipulative approaches aimed at localising and resolving somatic dysfunction and enhancing homeostatic mechanisms, as well as structure—function interrelationships.

Specific diagnostic and therapeutic strategies are employed in this process to ensure global patient care (Kuchera, 2007). The aim of osteopathic diagnostic palpation is to identify and resolve somatic dysfunctions, defined as any "impaired or altered skeletal, arthrodiastal, and/or myofascial function", related to neural and/or vascular elements, that might underlie pathophysiologic conditions (Allen, 1993). The observational and palpatory features of a somatic dysfunction include sensitivity to palpation, tissue texture changes (Fryer, 2003), positional (DiGiovanna et al., 2004) and/or functional (Greenman, 1989) asymmetry and restricted motion (Ward, 2003). If undiagnosed, somatic dysfunctions may result in persistent pain, either locally or at distant sites linked through compensatory mechanisms (Kuchera and Kuchera, 1994a). Depending upon the type, duration and amount of the load/stress as it occurs in dysfunctional patterns, fascia may respond in both a plastic and an elastic manner (Greenman, 1989), by reorganising itself along the lines of tension imposed or expressed in the body at both molecular (Mosler et al., 1985) and macroscopic level (Sasaki and Odajima, 1996). Known physiological connective tissue responses, following functional strain or mechanical stress through its collagen bundles, involve fibroblast mechano chemical transduction, modulation of gene expression patterns (Chiquet, 1999), together with inflammatory and tissue remodelling processes (Dodd et al., 2006; Langevin and Sherman, 2007) of the collagenous matrix (Grinnell, 2008; Swartz et al., 2001). Pain-related fear may then lead to decreased mobility. Consequent inflammation and nervous system sensitisation may follow, leading to further decreased mobility and connective tissue remodelling. Changes in collagen fibre density and orientation may take place until cross links develop between collagen fibres at the nodal points of fascial bands (Fratzl, 2008). In this fashion, resultant changes in tissue viscoelastic properties may occur, together with a change in the colloidal consistency of the ground substance in the fascia to a more solid

state (Cummings and Tillman, 1992). An US-based comparison of subcutaneous and perimuscular connective tissues forming the superficial and deep TLF, showed a 25% greater perimuscular connective tissue thickness and echogenicity in people with LBP, who expressed less relative tissue motion between the deep and superficial connective tissue of the back than the pain-free control group (Langevin et al., 2009).

Matrix remodelling and fascial contraction do not seem to be the only possible factors influencing fascial stiffening. Fascial strain hardening may also be related to stiffening of the dense fibrous connective tissue, due to a super-compensation effect, characterised by enhanced matrix hydration (Schleip et al., 2012). Such phenomenon occurs during the resting phase following a sustained static stretch, applied for 15 min to animal fascial tissue in vitro; whereas an extrusion of water with a related temporary reduction of stiffness was observed during stretch. Possibly, a similar behaviour may occur in a human fascial tissue presenting with somatic dysfunction and related abnormal tension.

Application of OFM and clinical evidences

Different osteopathic fascial techniques have been described, mainly at a post-operative stage (Schwartz, 1986; Stiles, 1976), proposing Osteopathic Manipulative Treatment (OMT) as an adjunctive treatment for patients with conditions such as post-operative ileus (Crow and Gorodinsky, 2009), congestive heart failure (Dickey, 1989), coronary artery disease (O-Yurvati et al., 2005), pancreatitis (Radjeski et al., 1998) and pneumonia (Noll et al., 2000). The results of these studies showed OMT as being easily implemented and cost-effective: the relief of acute pain allowed shorter hospital stays, early ambulation, decreased post-operative morbidity and mortality as well as increased patient satisfaction (Noll et al., 2000). However, with regards to being cost-effective OMT studies, *"measures are somewhat imprecise proxies for actual direct and indirect cost data. These variables are sometimes called "imputed costs" because they are not actual expenditures or costs; rather, a dollar value is imputed to them"* (Gamber et al., 2005).

The positive effect of OMT on leukocyte count (Hodge et al., 2007; Mesina et al., 1998) and IgA levels in highly stressed people suggests that OMT may have therapeutic preventive effects on both healthy and hospitalised patients, especially those experiencing high levels of emotional or physiological stress, and who are at risk of infections (Saggio et al., 2011). In particular, within the wide-ranging armamentarium available OMT and OFM incorporate fundamental manual tools for both diagnosis and treatment of somatic dysfunction and various clinical conditions. The efficacy of OFM has been demonstrated for acute ankle injuries (Eisenhart et al., 2003), carpal tunnel syndrome (Sucher, 1993), Dupuytren contracture (Sampson et al., 2011), infantile obstructive apnea (Vandenplas et al., 2008), otitis media (Zaphiris et al., 2004), peripheral artery disease (Lombardini et al., 2009), chronic asthma (Bockenbauer et al., 2002), headaches (Anderson and Seniscal, 2006), depression (Plotkin et al., 2001), fibromyalgia (Gamber et al., 2002), some symptoms of multiple sclerosis (Yates et al., 2002) and sexual dysfunction (Martin, 2004). Finally, OMT seems to be generally

effective for LBP (Licciardone et al., 2005; Seffinger et al., 2010), specifically for dysmenorrhea-related LBP (Boesler et al., 1993), and for LBP during the third trimester of pregnancy (Licciardone et al., 2010); in some studies OMT has been shown to be extremely effective in reducing or eliminating persistent LBP (Kuchera and McPartland, 2003) or acute LBP (Clark et al., 2009); in others as an effective adjunct therapy for LBP, regardless of whether or not pain radiates into the lower extremities (Blomberg, 1993); generally requiring less medication than standard medical therapies (Andersson et al., 1999); with physical and psychological positive outcomes in primary care with little extra cost (Williams et al., 2003). A study by the author has investigated whether pain patterns in patients suffering from non-specific LBP may vary together with the kidney range of mobility after OFM is applied in situ (Tozzi et al., 2012). It has been demonstrated, through US screening, that people with non-specific LBP present with a significantly reduced range of right kidney mobility, without a frank organic spinal or renal pathology, compared with that measured in asymptomatic subjects. The results have shown that application of OFM significantly improves kidney mobility and reduces pain perception over the short term, leading to a plausible fascial involvement in both renal mobility and LBP, as well as to a possible correlation between renal mobility and LBP.

The majority of these studies were conducted on small population samples, or as pilot studies, or single clinical case studies, usually with a short follow up to validate results over time, and sometimes with a poor control design. It also needs to be considered that, apart from the possible clinical results obtained in these trials, OMT may be generally perceived by participants as a more credible treatment than many control procedures (Licciardone and Russo, 2006). Treatment credibility may have interacted with subject expectations and study design, especially when people are asked to self-report data. In addition, because of the need of standardisation, OMT was applied mostly in terms of rigid protocols, rather than in the form of treatments tailored to patient's need, as would most usually be the case in osteopathic clinical practice.

No injuries have been reported in the literature as attributed to indirect or fascial techniques (Vick et al., 1996). However, OFM may result in a post-treatment myalgic flare, within the first 12 h after treatment. Usually lasting only a few hours, patients have described this flare as being similar in character and intensity to the muscle pain after a vigorous workout (Ward, 2003).

Osteopathic principles applied to fascia

In osteopathic practice there exist two main manual approaches to treatment of fascia, depending on whether the operator intends to address, or to move away from, the tissue barrier:

Direct Approach to Fascia: requires tissue's restrictions to be engaged and maintained until release is gained. Occasionally, as the affected tissue is brought against the functional barrier, a tridimensional compression or traction is applied and held (generally for 60–90 s) until tensions melt (Pilat, 2003). When the first barrier is released, the

procedure is repeated for consecutive barriers, adjusting the compressional force according to each barrier's vectors, up to a point when a release is felt. Pressure is reduced when there is any increase in pain. This is variously known as myofascial release, or myofascial induction.

Indirect Approach to Fascia: requires the exaggeration of the pattern of dysfunctional tissue motion, bringing the restricted fascial tissue into its position of 'ease' (balanced tension), maintaining it until tensional forces relax (Ward, 2003).

Osteopathic approaches to fascia

Within a wide-ranging armamentarium, the most used manual fascial techniques in osteopathic practice are outlined below and shown on Table 1. This list should be considered as partial with some of those listed being known by different names.

Myofascial Release: is defined by Manheim (2001) as the facilitation of mechanical, neural, and psycho physiological adaptive potential as interfaced via the myofascial system. It represents a widely known manual technique specifically for fascial tissues, designed to reduce adhesions, restore and/or optimise fascia's sliding mobility in both acute (Barnes, 1996) and chronic conditions (Martin, 2009; Walton, 2008). Some

studies have shown the efficacy of myofascial release to reduce pain (Barnes, 1990; Fernandez de las Penas et al., 2005), improve posture (LeBauer et al., 2008), and quality of life (Radjeski et al., 1998). It involves the application of a low load, long duration stretch, along the lines of maximal fascial restriction (Barnes, 1990). The practitioner palpates the latter and the pressure is applied directly to the skin, into the direction of restriction, until resistance (the tissue barrier) is felt. Once found, the collagenous barrier is engaged for 90–120 s, without sliding over the skin or forcing the tissue (Manheim, 2001), until the fascia complex starts to yield and a sensation of softening is achieved.

Still Technique (ST): is a method of manipulation that was first described by Andrew Taylor Still, founder of Osteopathic Medicine. Ashmore (1915), McConnell (1900) and Sutherland (1967) wrote descriptions of one method Dr Still used, called 'exaggeration of the lesion', that allowed "ligaments to draw the articulations back into normal relationship" (Sutherland, 1967). The principle of ST consists of manipulating the dysfunctional tissues to further increase the degree of malposition, into a position of 'ease', until relaxation occurs. Following this a vector of force (compression or traction) of less than 5 pounds (2 kg) is applied in a manner that forces the tissues through the initial restriction barrier, back towards a normal position

Table 1 The most used manual fascial techniques in osteopathic practice are shown on this table, together with their respective procedure of application. This list should be considered as partial with some of those listed being known by different names.

Osteopathic approaches to fascia		
Modality		Application
Myofascial release	Direct	It involves the application of a low load, long duration stretch, along the lines of maximal fascial restriction. The practitioner palpates the latter and the pressure is applied directly to the skin, into the direction of restriction, until the tissue barrier is felt. Once found, the collagenous barrier is engaged for 90–120 s, without sliding over the skin or forcing the tissue, until the fascia complex starts to yield
Still technique	Combined	It requires firstly determination of position of ease for the fascial element that is restricted; secondly the introduction and maintenance of a compressive force into the tissue; and finally the application of force to follow the tissue as it unwinds along its wandering pathway toward, and through, the position of initial restriction
Balanced lig. tension release	Indirect	It requires an initial disengagement of the joint, followed by exaggeration of the dysfunctional vectors of the somatic dysfunction until a ligamentous tensional compromise is achieved. This point is maintained, while the tensional and neurological information is elaborated up to when a release is achieved
Fascial unwinding	Indirect	The operator engages the restricted tissues/joint by unfolding the whole pattern of dysfunctional vectors enclosed in the inherent fascial motion. A shearing, torsional or rotational component may arise in a complex three-dimensional pattern that needs to be acknowledged, amplified and unwound until release is felt
Harmonic technique	Indirect or direct	The frequency of the driving force produces a maximal effect when it equals the natural frequency of the segment. When this occurs, a constant amplitude of oscillation is maintained and the resonance frequency of the system is achieved. When applied to fascia with a functional intent, ease is pursued with minimal oscillatory traction. If applied with direct intention, bind is engaged and ranges of movement are improved by oscillatory traction until a tensional equilibrium is restored
Strain& counterstrain	Indirect	It implies the taking of the tissues to the point of ease, monitoring this process by means of assessing tender points. By palpating the tender point and asking the patient for feedback regarding the level of tenderness, the operator can accurately localise the exact position of ease as the tenderness disappears. The patient is then held in this position for about 90 s, and then very slowly returned back a neutral position

(Hazzard, 1905). By using the natural recoil of the structures from their exaggerated position, a benefit is gained, and this motion is thought to release tension and loosen adhesions. When applied to fascia, the ST requires firstly determination of the position of ease for the fascial element that is restricted; secondly the introduction and maintenance of a compressive force into the tissue; and finally the application of force to follow the tissue as it unwinds along its path toward, and through, the position of initial restriction (Van Buskirk, 2006).

Balanced Ligamentous Tension Release: described by WG Sutherland (1949), this approach proposes that all joints in the body are balanced ligamentous articular mechanisms. The ligaments provide proprioceptive information that guides the motion of the articular components as well as the muscle response for positioning the joint (DiGiovanna et al., 2004). The general procedure requires an initial disengagement of the joint, this is followed by exaggeration of the dysfunctional vectors of the somatic dysfunction (moulding phase) until a ligamentous tensional compromise is achieved (a balance point). *"The point of balanced tension is a sensation of contrast of freedom and restriction of mobility. The skill is holding the tissues at the meeting point of this contrast"* (Magoun, 1976). This point is maintained, while the tensional and neurological information is elaborated (refining stage) up to when a point of tensional and proprioceptive quietness takes place (still point). Eventually, a change in the palpatory properties of the structure being treated occurs (releasing stage), such as a change in skin temperature, fascial/muscle tension, articular range of mobility.

Fascial Unwinding: is a commonly used technique in osteopathic practice (Johnson and Kurtz, 2003; Ward, 2003), aimed at releasing fascial restrictions and restoration of tissue mobility and function. It comprises a dynamic functional indirect technique applied usually to the entire myofascial-articular complex: the operator engages the restricted tissues/joint by unfolding the whole pattern of dysfunctional vectors enclosed in the inherent fascial motion. A shearing, torsional or rotational component may develop in a complex three-dimensional pattern that needs to be supported, amplified and unwound, until release is perceived (Ward, 2003).

Harmonic Technique: Harmonic Technique is a system that works through the oscillatory nature of the body's tissues *"bringing on a state of resonance within these tissues"* (Lederman, 1990). An endogenous rhythmic property of the body, primarily driven via the nervous system and its dynamic plastic property, and also known as tonic vibratory reflex, seems to be involved (Comeaux, 2008). The frequency of the driving force produces a maximal effect when it equals the natural frequency of the segment. When this occurs, a constant amplitude of oscillation is maintained and the resonance frequency of the system is achieved (Sernay, 1975). The dampening of such inherent body oscillatory property indicates a possible area of dysfunction. Hold, direction, amplitude, speed and rhythm are all essential components of this approach that may work as both a diagnostic and a therapeutic tool. When applied to fascia with a functional intent, ease is pursued with minimal oscillatory traction. If applied with direct intention, bind is engaged and ranges of movement are

improved by oscillatory traction until a tensional equilibrium is restored.

Strain & Counterstrain (SCS): introduced in the osteopathic field by Jones (1964), this approach is biomechanically indirect in its nature, and implies the taking of the patient (or local tissues) to the point of ease, monitoring this process by means of assessing tender points. Jones believed the mechanisms involved in SCS to be mainly neuro physiological (Patterson, 2002). The tender points exhibited in the body, in relation to dysfunction, are small areas, usually about 1 cm across, of oedematous and tense muscle or fascia that are tender to palpation. By palpating the tender point and asking the patient for feedback regarding the level of tenderness, the operator can accurately localise the exact position of ease as the tenderness disappears. The patient is then held in this position for about 90 s, and then very slowly returned back a neutral position, particularly slowly in the first 15° of movement (Jones, 1981). This original approach has been modified by subsequent practitioners such as Chaitow (2007), D'Ambrogio and Roth (1997) and Deig (2001). According to D'Ambrogio and Roth (1997) the range within which a state of balancing forces is palpated through the dysfunctional tissues is quite limited to between 2 and 3°.

Note: All the techniques listed above, as with most osteopathic techniques, were developed by subjective experience arising from individual clinical practice. Different osteopaths have suggested ideal protocols for application for their own therapeutic method, perceived as the most efficient by themselves, in the attempt to reproduce specific tissue responses and expected clinical results. However, such protocols have often been poorly validated in controlled clinical settings, leaving the scientific validity and efficacy in doubt. Yet, these same protocols have been passed down through generations of osteopaths, often in a dogmatic fashion, despite their intrinsic subjective nature and their explanations being mostly based upon theoretical conjectures. Even where some of these treatment protocols have been revisited by others in the light of updated paradigms of scientific and clinical knowledge, an osteopath often finds him/herself in a controversial position, when it comes to choosing which technique is most suitable for his/her patient. In fact, within the osteopathic community, specific treatment criteria for a given patient/condition are not yet established. Certainly there are guidelines, but the process is left mainly to an individuals 'educated guess' based on personal experience and clinical knowledge. A better understanding of the physiology and mechanism behind a techniques' effectiveness, as well as a sound clinical evidence base would certainly help the profession to provide clarity with modalities of application. However, in the author's opinion, osteopathic medicine should never be reduced into a series of sterile prescriptions for each given complaint, which would undermine the inherent principles of holistic medicine as well as the aim of health promotion in an individual rather than aiming to cure disease.

Possible physiological effects of OFM

Assuming that the choice of the therapeutic strategy and objective is always based on each patient's unique clinical

presentation, OFM may play a beneficial role on modulation of such pain patterns, by influencing certain circulatory, respiratory, neuro endocrine and autonomic mechanisms (Kuchera, 2007), summarised on Table 2.

For instance, during application of the Still Technique, the practitioner starts treatment by unloading the tissue, consequently decreasing neural inputs and physical stresses through the tissue. Van Buskirk (2006) hypothesises that during the positioning of ease, the application of the force vector will unload the muscle spindles while possibly loading Golgi tendon organs. This may change the pattern of sensory input to the facilitated spinal cord area, quieting the nociceptors and diminishing the release of substance P that has been maintaining local oedema. The local and spinal cord level autonomic reflexes will be stilled, particularly the sympathetic drive which may have encouraged vasoconstriction and diminished lymphatic flow (Van Buskirk, 2006). This may restore the normal pumping action of the muscles and fascial movement, improving venous and lymphatic flow through the tissue. As blood and lymphatic flows are optimised, oxygenation would be

restored to normal levels, nutrients brought to the tissue cells at an appropriate concentration, and toxins and inflammatory products removed.

The respiratory–circulatory osteopathic model, proposed by Zink and Lawson (1979) implies the opening of fascial pathways leading to a reduction of oedema and associated peripheral biochemical molecules linked to nociception.

A modelling of indirect OMT, at a cellular level, has been applied as a cyclical unequal strain for 60 s to human fibroblasts, after which they were repetitively strained for 8 h in a two dimensional tissue culture matrix. A subsequent 60 s introduction of positional release (creating an 'ease' position) produced beneficial effects on fibroblast morphology and actin stress fibre architecture of the tissue sample. This reversed the inflammatory effects in cells (Meltzer and Standley, 2007) suggesting that fibroblast proliferation and expression/secretion of anti-inflammatory interleukins may contribute to the clinical efficacy of indirect OMT. Furthermore, manual loading of fascia as in various manual treatments may cause changes

Table 2 The main possible physiological effects of OFM are listed, together with a short description of plausible mechanisms leading to tissue change during or after treatment.

Possible physiological effects of OFM

Neuromuscular	By unloading the tissue, indirect OFM may decrease neural inputs and physical stresses through the tissue. During the positioning of ease, the application of the force vector may unload the muscle spindles while possibly loading Golgi tendon organs. This may change the pattern of sensory input to the facilitated spinal cord area, quieting the nociceptors and diminishing the release of substance P that has been maintaining local oedema
Structural changes	Structure of the collagen matrix in the dermis can be changed by manual therapy in areas of the body where patients experience discomfort. The changes mirror the differences in tension, softness and regularity, that can be palpated before and after treatment. Myofascial release has shown an increase of fascial thickness, with a persistence of such changes for at least 24 h
Viscosity changes	OFM may lead to a transformation of the ground substance from its densified state (gel) to more fluid (sol) state. This change in viscosity seems to increase the production of hyaluronic acid, together with the flow within the fascial tissue: to improve drainage of inflammatory mediators and metabolic wastes; to decrease chemical irritation of the ANS endings, and nociceptive stimuli to somatic endings, therefore to reset aberrant somato–visceral or visceros–somatic reflexes
Fluid flow	Increased interstitial fluid flow stimulates fibroblast proliferation, fibroblast to myofibroblast differentiation, and collagen production/alignment, leading to the speculation that OFM may play a role in fibrogenesis and fascial repair. As blood and lymphatic flows are optimised, oxygenation would be restored to normal levels, nutrients brought to the tissue cells at an appropriate concentration, toxins and inflammatory products removed
Cellular response	Manual loading of fascia may cause changes through activation of fibroblasts response and the different receptors present in the fascial tissue, leading to corrections of fascial hypertonicity and/or abnormal tissue collagen crosslinking
Endocannabinoids	OFM may lead to an anandamide effect on the eCB system: the activation of the eCB system through OFM diminishes nociception and pain, reduces inflammation in myofascial tissues and plays a role in fascial reorganisation
Sympathetics	Pain reduction following OFM has also been linked to modulation of hypersympathetonia and parasympathetic tone with an improvement in a variety of visceral and somatic features, as demonstrated by hemodynamic functions, heart rate variability and peripheral artery disease
Respiration	Respiratory contribution during OFM may play a role in myofascial relaxation and improvement in articular/tissue mobility
Oscillations	A slow rhythmic pendular swing has been shown to cause an inhibitory effect on vestibular nuclei, resulting in muscle relaxation, as well as in a significant change in motoneuron excitability. Oscillations may also promote inter-compartmental fluid flow through hydraulic mechanisms as well as a possible modulating effect on the pain gate mechanism within the spinal cord

through activation of fibroblast responses (Eagan et al., 2007) and the different receptors present in the fascial tissue (Lundon, 2007; Schleip, 2003b), leading to corrections of fascial hypertonicity and/or abnormal tissue collagen crosslinking.

Structural changes

A US scan-based study has shown that the structure of the collagen matrix in the dermis can be changed by manual therapy (connective tissue massage) in areas of the body where patients experience discomfort such as chronic pain (Pohl, 2010) (Fig. 5). The changes mirror the differences in tension, softness and regularity, that can be palpated before and after treatment. In particular, myofascial release of the thoraco–lumbar fascia (TLF), in people with chronic LBP, has shown a US scanned increase of its thickness, with a persistence of such changes for at least 24 h (Blanquet et al., 2010). Fascial techniques involving firm frictional compression, applied to the TLF of people with LBP, seems to require different periods to modify apparent fascial density/fibrosis, depending on characteristics of the subjects and duration of the symptoms: in those subjects with symptoms present for less than 3 months (sub-acute) the mean time to gain a tissue release was less (2.58 min) with respect to that observed in chronic patients (3.29 min)

(Ercole et al., 2010). Nevertheless, with regards to plantar fascia and fascia lata, a three-dimensional mathematical model for deformation of human fasciae has suggested that palpable sensations of tissue release following manual therapy cannot be due to deformations produced in such firm tissues (Chaudhry et al., 2008).

What is clear is that the myofascial system seems to respond not only locally to the tissue where the treatment is applied but also in segments connected to the treated area: a cervical myofascial induction technique to the ligamentum nuchae in healthy subjects resulted in an increase of all cervical ranges of motion, apart from rotation (Saiz-Llamas et al., 2009). This phenomenon may indicate an electrical and/or a mechanical response of the myofascial structures indirectly connected with the area being treated (Kassolik et al., 2009). The latest research suggests that even muscle performance may be indirectly improved, following a manual release of the surrounding fascial structures, possibly through the tension redistribution of the fascial net itself: US scanning in people without LBP has shown an increase in the lateral sliding of TLF–transversus abdominis junction following application of manual fascial release for just 1 min at the anterior and posterior region of the transversus abdominis. The study showed that the total length of the transversus abdominis muscle may be further shortened after release, suggesting

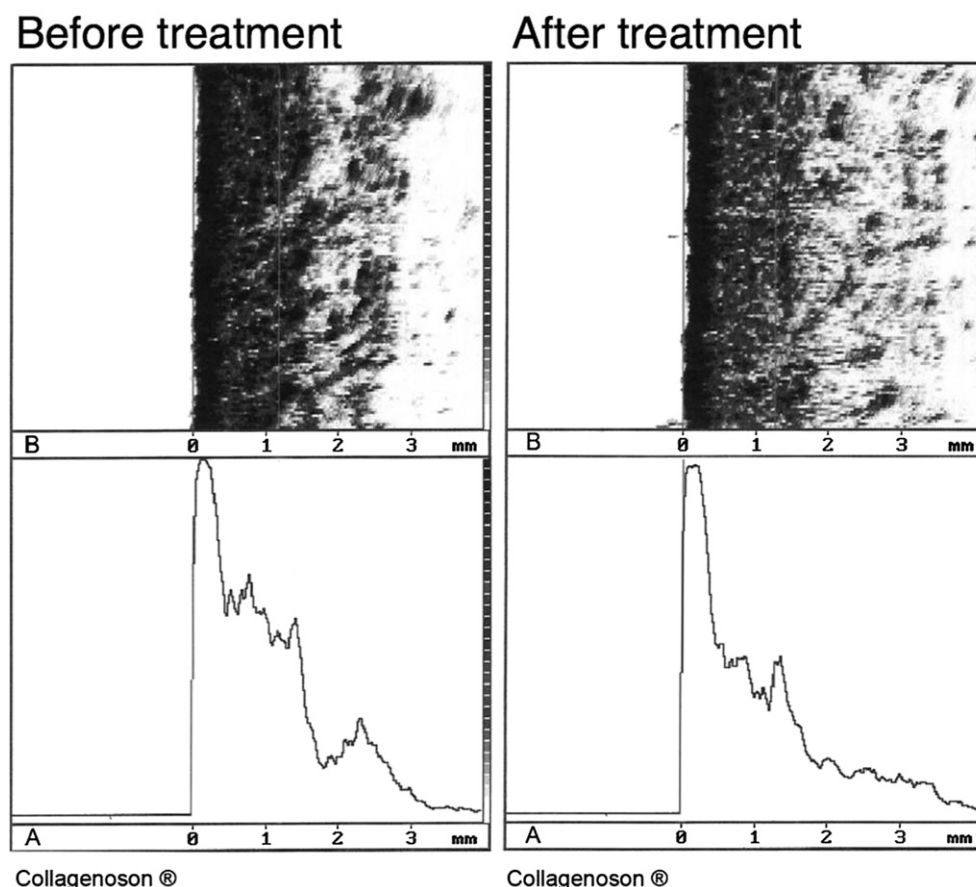


Figure 5 In a 64-year-old female, with pain felt on top of the left foot, the distribution of bundles of collagen fibrils, before and after (connective tissue massage) treatment, are shown in both a B = brightness scan and a diagram A = amplitude scan. The latter shows the curve of summed collagen density in the various layers of the skin, from epidermis (on the left side of the pictures) to subcutis (on the right side of the pictures). Reproduced from Pohl (2010) with permission.

that it may increase muscle activation and contractibility (Chen and Wang, 2010).

Viscosity changes and fluid flow

Pain patterns in people suffering from non-specific neck pain (NP) or low back pain (LBP) have been shown to improve, together with an improvement of the range and quality of surrounding fascial sliding motion, after OFM, applied in situ (Tozzi et al., 2011). Research suggests that increase in sliding of the tissue layers, together with a decrease in pain following manual fascial work, may be the result of a transformation of the ground substance from its densified state (gel) to more fluid (sol) state (Day et al., 2009; Pedrelli et al., 2009). This change in viscosity seems to increase the production of hyaluronic acid, together with the flow within the fascial tissue: to improve drainage of inflammatory mediators and metabolic wastes (Schultz and Feltis, 1996); to decrease chemical irritation of the autonomic nervous system (ANS) endings, and nociceptive stimuli to somatic endings (Lund et al., 2002; Mense, 1983), therefore to reset aberrant somato-visceral and/or viscerosomatic reflexes (Bandein, 1949; Mannino, 1979).

In addition, such increased interstitial fluid flow stimulates fibroblast proliferation, fibroblast to myofibroblast differentiation, and collagen production/alignment (Hinz et al., 2004), leading to the speculation that OFM may play a role in fibrogenesis and fascial repair by promoting interstitial fluid flow (Chila, 2010). OFM may produce these clinical outcomes due to enhancement of cytokine pools (Chila, 2010) from actively proliferating fascial fibroblasts and correction of dysfunctional matrix crosslinking (Gupta et al., 1998). Furthermore, the release of cytokines seems to cause selective fibroblast death from injury sites, together with removal of fibrotic materials (Dodd et al., 2006). This phenomenon may have contributed to the reduction of the range of mobility and to development of pain. Such fascia-derived cytokines may be delivered systemically, even in distant sites to that treated with OFM, via intrafascial blood flow (Bhattacharya et al., 2005), possibly reducing oedema, increasing range of mobility and decreasing pain (Meltzer and Standley, 2007), even at sites distant to those where the technique is applied.

Cellular responses

Beneficial strain, such as that applied during a therapeutic session, may be sensed at the cellular level by integrin-independent alterations in membrane ion conductances (Yang et al., 2004) and by mechano-gated calcium-ion channels (Gupta et al., 1998), which activation appears to be dependent upon an intact actin cytoskeleton (Mohanty and Li, 2002; Wang et al., 2005). The structural framework found at a cellular level allows forces to be transmitted within the cell, so that a mechanical stimulus to the cell surface may evoke a reaction and immediate changes in the cytoplasm and nucleus (Chen and Ingber, 1999). In addition to strain frequency, magnitude and duration, strain direction differentially regulates fibroblast and myofibroblast growth, ion conductances and gene expression (Berry et al., 2003; Hornberger et al., 2005), responding accordingly with

differential stretch-activated calcium channel signalling (Kamkin et al., 2003; Xu et al., 2000). Equi-biaxial strain of tendon fibroblasts for 2 h resulting in inhibition of IL-1B expression showing that fibroblast strain may set in motion a cascade of events that attenuate proinflammatory while at the same time stimulating anti-inflammatory signalling pathways (Tsuzaki et al., 2003), therefore influencing pain perception.

Endocannabinoids

In addition, following the application of OFM, reduced pain patterns may be related to the OFM-related anandamide effect on the endocannabinoid (eCB) system (McPartland et al., 2005): an endorphin system constituted of cell membrane receptors, endogenous ligands and ligand-metabolising enzymes. This system affects fibroblast remodelling and dampens cartilage destruction. Therefore, the activation of the eCB system through myofascial manipulation diminishes nociception and pain, reduces inflammation in myofascial tissues and plays a role in fascial reorganisation (McPartland et al., 2005). This has been shown to occur specifically in subjects with chronic LBP following OMT (Degenhardt et al., 2007), who identified a significant alteration in the concentration of several circulatory pain biomarkers (beta-endorphin, 5-hydroxytryptamine, 5-hydroxyindoleacetic acid, arachidonylethanolamide, N-palmitoylethanolamide). In particular, there was a twofold increase in N-palmitoylethanolamide at 30 min and at 24 h post-treatment with respect to that in the control subjects, demonstrating a correlation with a decrease in stress occurred from baseline to day 5 post-treatment.

Changes in sympathetic tone

Pain reduction has also been linked to modulation of hyper-sympathotonia by applied OMT, with an improvement in a variety of visceral and somatic features (Kuchera and Kuchera, 1994b; Van Buskirk, 1990), as demonstrated by hemodynamic functions (Huard, 2005; Rivers et al., 2008) heart rate variability (Henley et al., 2008) and peripheral artery disease (Lombardini et al., 2009) following osteopathic fascial work. Parasympathetic tone may also have an influence as its up regulation, following manual therapy, has been reported to evoke an increase in heart rate variability (Vagedes et al., 2009), together with an influence on blood shear rate and blood flow turbulence in particular (Querè et al., 2009).

Respiration and oscillation

It is not uncommon to request patient forced respiration at some point during a fascial treatment. Such respiratory contribution may play a role in myofascial relaxation and improvement in mobility. Cummings and Howell (1990) have demonstrated the effects of respiration on myofascial tension whereas Kisselkova (1976) reported that resting EMG activity of non-respiratory muscles were cycled with respiration, suggesting that they receive input from the respiratory centres. The influence of respiration on the musculo skeletal system seems to be plausible, whereas its frequency certainly has the ability to be synchronised and

to interact with oscillations in blood pressure (Daly, 1986), heart rate (Song and Lehrer, 2003), lymphatic system (Gashev, 2002) and brain waves (Busek and Kemlink, 2005), together with being amplified due to resonance effects between these systems. "Oscillations in single systems and synchronization between oscillating systems help physiological control systems to maintain homeostasis and appropriate and rapid responsiveness to the continual changing needs of the body" (Courtney, 2009). Vibration or oscillation has been used as a component of diagnosis and treatment in various forms since the beginning of osteopathy. Sutherland (1950) suggested the benefits of vibration applied to the lymphatics, whereas Mitchell (1995) proposed the addition of vibrations to counteract the myotactic reflex in hypertonic muscles. Fulford invented a percussion vibratory treatment to enhance "freedom of function on all levels, using the energy levels of the body as the vehicle for diagnosis and treatment" (Comeux, 2002). When applying this principle on a therapeutic level various physiological effects can be evoked in the body. A slow rhythmic pendular swing has been shown to cause an inhibitory effect on vestibular nuclei, resulting in muscle relaxation, as shown in hyperactive children helping to reduce their excitatory state (Ayres, 1979) by inducing a psychogenic relaxation. Low frequency oscillation can induce muscle relaxation, provoking a significant change in motoneuron excitability (Hogbarth and Eklund, 1969; Newham and Lederman, 1997). It has also been demonstrated that both primary and secondary nerve endings are sensitive to vibration and sinusoidal oscillation (Walsh, 1971; Bach et al., 1983): during passive oscillation of a joint, a build up in amplitude of response has been observed, when oscillations near the resonant frequency was applied. Moreover, oscillatory forces appear to provoke a greater effect on spine mobility than structural manipulation (Keller and Colloca, 2002). Oscillations may also promote inter-compartmental fluid flow through hydraulic mechanisms (Lederman, 1997) as well as have a possible modulating effect on the pain gate mechanism within the spinal cord: it has been used in manipulative therapy to assist the resolution of inflammatory processes and to modulate pain perception (Wells, 1985).

Conclusion

It seems evident that various factors may interplay with myofascial function and its ability to respond to treatment. However the connective tissue may serve as a *trait d'union* of all these elements, potentially sustaining an integrated understanding of the whole picture: 'Since connective tissue plays an intimate role in the function of all other tissues, a complex connective tissue network system integrating whole body mechanical forces may coherently influence the function of all other physiological systems. Demonstrating the existence of such "meta-system" would therefore change our core understanding of physiology' (Langevin, 2006).

Conflict of interest statement for author

Paolo Tozzi. does not disclose any actual or potential conflict of interest including any financial, personal or other

relationships with other people or organisations within three years of beginning the submitted work that could inappropriately influence, or be perceived to influence, this work.

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